AGNIRATH

MECHANICAL MODULE

RECRUITMENT'26

APPLICATION FOR THE POSITION OF CREW MEMBER

AGNIRATH

GENERAL INSTRUCTIONS

- Try to answer as many questions as possible (All sections)
- You can use the help of any online resources to answer the questions.
- We have attached a few references (hyperlinks in questions) which will be helpful to solve the questions.
- Try to be creative but feasible with your ideas.
- Explain your ideas/strategy wherever possible.
- Be as concise and clear as possible.
- Clearly state any assumptions that you're taking in every question.
- Write your answers in points instead of paragraphs. Incase of any doubts, feel free to reach out to the respective leads.
- Save your answer script as a PDF file with the name <moduleacronym>_<rollnumber>_<name>.pdf
  Eg: ME_AE25xxxx_SARVS.pdf (ME for Mechanical Module)

Resources to refer

- Application submissions are to be made through Agnirath Website. Details of which will be shared on Whatsapp group later on.
- The deadline for submission is **19th April 2026**

AERODYNAMICS & COOLING

ALFIE
6383953512

RAGHAV
8778326165

VEHICLE DYNAMICS

ADITYA
6381938078

ARUDHRA
9150135170

STRUCTURES & COMPOSITES

SARVESH
6383904532

OMKAR
7842326994

SURESH
8790478236

FOUNDATION

AGNIRATH

ENERGY AUDIT

The Bridgestone World Solar Challenge is about 3000 kilometers long, which must be traversed entirely using solar energy. Teams are allowed to run the car from **08:00 to 17:00**, and the race lasts for about **5 days**.
Hence, energy becomes a very important resource, and the entire car must be designed with this in mind to minimize losses as much as possible.
Assume that the sun rises at 06:00 hrs and sets at 18:00 hrs, with peak incoming sunlight at 12:00 hrs. During the time outside the race hours, the car can charge in an ideal state. The car also charges throughout the race time. The system starts with a 1.6 kWh battery pack fully charged at the beginning of the race.
Since the entire race revolves around energy efficiency, it is essential to reduce energy wastage wherever possible.

Questions :

- Compute the minimum speed required to complete the race within the given time.
- Perform a basic energy audit, listing the various ways in which energy is gained and lost in a solar race car.
- Which energy losses/gains contribute the most? What can be done to increase energy gain and reduce energy loss?
- Formulate these energy gains and losses to derive the energy-neutral speed of the car.
- Represent the dissipative forces acting on the solar car in the form of a Free Body Diagram (FBD).
- The electrical subsystem of the car operates on a three-way bus between the solar array, battery, and motor. Represent the energy flow using a flow diagram.
    - State the three energy balance conditions the car undergoes.
    - Mention real-world instances where each condition occurs.

VEHICLE DYNAMICS

AGNIRATH

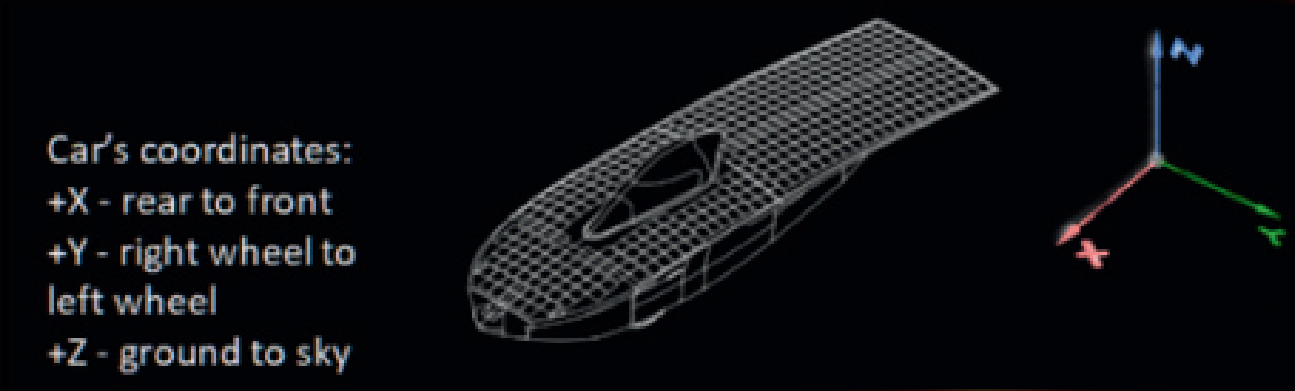
STABILITY ANALYSIS

Question 1.1

You may have noticed that Aagneya has only 3 wheels instead of 4.
Analyze the dynamic stability of 3 wheeled cars vs 4 wheeled cars, and determine which one is more stable. Why do you think Aagneya has 3 wheels, and explain why the tradeoff is acceptable. Also think about how VD is affected in both the cases.

Question 1.2

The teams which participate in the World Solar Challenge usually adopt one of 2 main car designs. These are the Bullet and the Catamaran designs.
Read about these 2 car configurations, and determine the stability characteristics (Roll stability, Load transfer, yaw stabilization). Think about the conditions in which the stability of the car is affected, and analyze the stability characteristics for all the case(s).
Also think about how the Vehicle Dynamics of the car is different between these 2 car designs.



These are the car's coordinate axes (for future questions)

VEHICLE DYNAMICS

AGNIRATH

CORNERING AND LATERAL FORCES

Question 2.1

In F1 races, you might have noticed that car does not always move as the driver wants it to. If driver wishes to take a sharp turn, the car might take a longer turn and vice versa.

Identify these 2 phenomena, and explain what they are and why they happen. Try to mathematically highlight when each of these conditions would occur.

Question 2.2

Being a 3 wheeled car, cornering is a concern for us. Thus, a lot of caution must be taken while taking a turn to prevent the car from slipping out of the turn.

Which tire property is cornering capability dependent on?
Explain what you understand by lateral forces and what it depends on.

Question 2.3

There is a car of mass 981kg (A normal 4 wheeled car), with a 2.5m wheelbase and a 1.2m trackwidth, which is trying to navigate a turn of radius 15m. The coefficient of friction (μ) between the tires and the road is 0.9.

The vehicle is in neutral steer, and is undergoing highspeed cornering, which induces some slip angle, causing the car to move along a circle with center on the line passing through the center of the wheelbase, and perpendicular to the wheelbase. It is seen that the car is JUST able to take this corner.

Assume that all tires are identical, and that the Lateral force is directly proportional to the slip angle. What is the cornering stiffness of each wheel?

(To solve this question, start by drawing the correct diagram. Then understand what you mean by JUST able to take a corner)

VEHICLE DYNAMICS

AGNIRATH

STEERING

Question 3.1

Steering plays an important role in controlling the overall efficiency of the car, as an inefficient steering system would lead to a lot of energy losses.

There are 2 main types of steering systems adopted - Ackermann steering system and the anti-Ackermann steering system

What are the advantages of each of these systems? Identify which of these two is used in Aagneya and back your answer with proper reasoning.

Question 3.2

What do you understand by the following terms -

- Kingpin axis inclination
- Caster
- Camber
- Toe
- Scrub Radius

What do you think is the range of scrub radius which should be used in Aagneya? Give proper reasoning for your choice.
Let's say that during track testing, we find out that a tire is cambered. What do you expect to observe on the tire if it runs for a considerable distance. How will the tires be different from other non-cambered tires after track testing?
If the tire has an initial toe instead of camber, how will the pattern be different?
After this, conclude by determining the ideal camber, and toe ranges for our car.

While building a car, it is important that every module, and every sub module in the team follow a common coordinate system, to eliminate any confusion during assembly.
We have attached the coordinate directions which we use in Team Agnirath at the end of Q1 of VD.

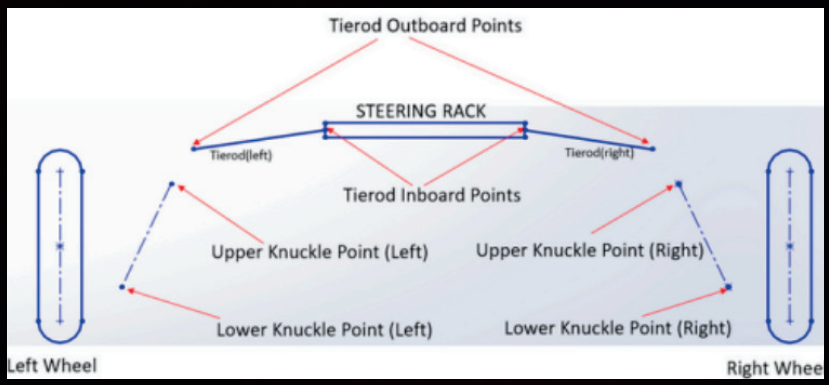
VEHICLE DYNAMICS

AGNIRATH

Question 3.3

Given below are the coordinates of our Steering system and a diagram to help you understand the position of each component. Understand the steering mechanism to the best of your ability, and proceed.

Component	Left Coordinate	Right Coordinate
Lower Knuckle	(0, 481, 156)	(0, -481, 156)
Upper knuckle	(-25, 349, 576)	(-25, -349, 576)
Tierod Inboard	(183, 249, 585)	(183, -249, 585)
Tierod Outboard	(157, 343, 580)	(157, -343, 580)
Front wheel center	(0, 600, 278.5)	(0, -600, 278.5)
Rear Wheel centre (-2600, 0, 278.5)		



The left wheel is turned to the left, from 0° to 20°, through an increment of 0.1° in every iteration. For each of these iterations:

- Find angle by which right wheel turns, if it were 100% Ackermann steering
- Find angle by which right wheel actually turns, using geometric constraints, and verify if the steering is 100% Ackermann or not.

If the actual steering is not 100% Ackermann plot the Ackermann Error vs the angle by which the left wheel is turned. Also tell why you think it's not at 100% Ackermann for the given coordinates (Not the numbers, but why haven't we kept it in Ackermann if it's not in 100% Ackermann).

Ackermann Error is defined as the difference between the angle turned by the right wheel, and its expected turning angle if it were 100% Ackermann.

You may use Python or MATLAB to do so.

VEHICLE DYNAMICS

AGNIRATH

BRAKING

Question 4.1

Braking is an essential part of any car, and hence we must make sure that it is as efficient as possible. There are 2 main types of brake rotors. They are the solid and the floating brake rotors. Read up about both of these, compare them, determine which one of these is better, and justify your answer.

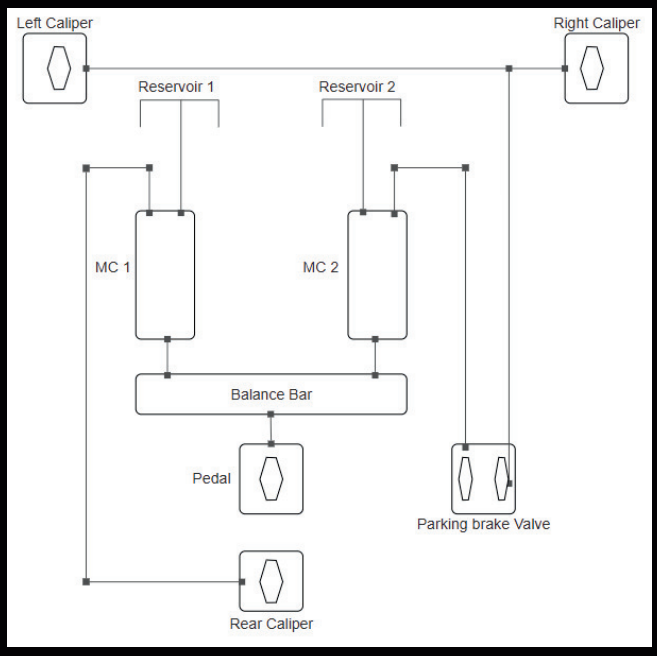
Question 4.2

A schematic diagram of the brake lines and brake design in Aagneya is given below. Explain the schematic in your own words.

Necessary Data for this Question:

- Pedal Ratio = 4:1
- Bias at Balance Bar = 2 : 1 (Front : rear)
- Piston Area in MC = 200mm$_2$
- Brake Pads Area = 750mm$_2$
- μ_s between tire and ground = 0.7
- μ_k between tire and ground = 0.65
- μ_k between caliper and rotor = 0.2
- Moment of inertial of wheel = 0.5kgm^2
- Radius of wheel = 278.5mm

You may assume that the center of the brake pads is at a distance of 100mm from the center of the brake rotor



The car is of mass 300kg, and is moving at a speed of 60kmph. The driver pushed the brake pedal with a force of 700N.

- How much time does it take for the wheel to stop
- How much time does it take for the car to stop

(You may neglect dynamic load transfer for this question)

Are these timings different and if so, why? Also, using the above data, calculate the brake bias. After considering all factors (list the factors you have considered), determine if it is good or bad. If it is bad, how will you change the brake bias?

**The MC in the diagram is known as the Master Cylinder*

VEHICLE DYNAMICS

AGNIRATH

TRACK TESTING

Question 5

One of the major contributors to energy loss is Rolling Resistance. This is a property of the tire. However, the tire manufacturers do not give us this data, and hence we need to test the car in order to calculate this parameter.

Another parameter we do not know is the Coefficient of Drag (Cd) which is usually calculated and expressed as CdA where A is the area.

Start by explaining what you understand by Rolling Resistance. How is it different from Rolling friction? Research about different tests we could do during track testing to determine unknowns such as the Coefficient of Rolling Resistance.

We have some data from one of our track tests in which we performed the coast down test of our car which weighs 300 Kgs. Derive the expression to calculate the Coefficient of Rolling Resistance through this test. While deriving the expression, consider the coefficient of lift as well.

time (s)	velocity of vehicle wrt ground(kmph)
0	20
100	17.08381587
200	14.39029938
300	11.87103749
400	9.486990057
500	7.20575642
600	4.99963578
700	2.844170792
800	0.717000946

You may assume that the ratio of the coefficient of drag to the coefficient of lift is greater than the coefficient of rolling resistance.

While calculating the Coefficient of Rolling Resistance using the given data, you may assume that the Coefficient of Lift is 0, and you may also assume that the wind speed is 0.

STRUCTURES AND COMPOSITES

AGNIRATH

STRUCTURES

The chassis is the fundamental structural frame that supports all major components of a car while ensuring the driver's safety from impacts and external loads. All loads acting on the vehicle ultimately transfer onto the chassis, which are then experienced by the occupants.

Question 1.1

One important aspect of a chassis is its stiffness. It is essential for a chassis to be stiff and rigid but why? One way to measure the stiffness is through its torsional rigidity. How can you quantify the torsional rigidity of an object? State the assumptions and idealizations involved in this process. Do you think the same assumptions can be applied to FRP materials?

Question 1.2

A chassis serves as the primary framework that supports all the loads acting on a vehicle. So, what are the loads that affect an automobile chassis? These loads can be classified based on time as Static loads and Dynamic loads. In dynamic loading, there are impact and fatigue loads.

Which parts of the chassis do you think will experience fatigue loads? Are composite materials well suited for handling fatigue loads?

Question 1.3

The occupant's cell, where the occupant is seated, must be able to withstand specific impact loads. In the architecture of a commercial vehicle, how are these impact loads managed? Which part of the chassis is responsible for bearing such impact loads? Suggest some methods to reduce the impact loads experienced by the occupant?

STRUCTURES AND COMPOSITES

AGNIRATH

COMPOSITES - DESIGNING

Our car should be as light as possible, without compromising on structural integrity. We use Fiber Reinforced Composites to achieve this. This consists of a matrix system reinforced by fiber based reinforcements. By tailoring the composition and orientation of the fiber, we engineer the desired properties.

Question 2.1

Team Agnirath participates in various International Solar Challenges. The competition is more about endurance and reliability than speed. Successfully completing and winning the challenge requires a well-designed, robust car.

Do you think making car only out of composites is viable? What considerations should be taken into account? Do a basic SWOT analysis on composites.

Question 2.2

Consider a Unidirectional Composite material subjected to purely longitudinal loading. Explain how the applied load is shared between the fibers and the matrix. How does the stiffness of the overall composite material depend on its constituents? Derive an expression for it, by quantifying stiffness.

Critical Volume Fraction is the minimum volume fraction of reinforcement that ensures strengthening of the matrix. State necessary assumptions and derive the expression for it.

Question 2.3

Fiber Reinforced Composites are highly anisotropic in nature. Majority of the physical properties vary with direction. What are the different types of fiber weave architectures used in composite materials? For each type, discuss the advantages and limitations in aspect of mechanical performance and manufacturability.

STRUCTURES AND COMPOSITES

AGNIRATH

Consider the following components manufactured using carbon fiber reinforced polymer (CFRP):

- A table tennis racket
- A drive shaft

From a materials engineering perspective, recommend the most suitable fiber architecture (weave pattern or fiber orientation) for each component and justify your selection based on loading conditions and performance requirements.

Question 2.4

In order to perfectly tailor our stacking sequence based on our requirements, we need to understand how the elastic constants vary with the orientation of fibers. Obtain the values for the elastic constants with the reference, when the ply axes are oriented at 30 degrees, 45 degrees and 60 degrees with the reference. State the necessary assumptions

E_{Warp} = 59.16 GPa
E_{Weft} = 59.16 GPa
Poisson's Ratio (along the ply) = 0.04
Shear Modulus (along the ply) = 3.3 GPa

Question 2.5

A car chassis is to be constructed using CFRP sandwich panels. Each panel consists of PW400 carbon fiber sheets and a H100 PVC foam core. Each carbon fiber ply is of 0.4mm thickness, and the foam core is of 5mm thickness.

A rectangular specimen is extracted from the panel such that its width is equal to twice the total thickness of the specimen and span-to-depth ratio is 16. This specimen is subjected to a load of 50 Kgf applied at the mid-span.

The laminate stacking sequence is: [0°/45°/0°/45°/0°/core /0°/45°/0°/45°/0°]

STRUCTURES AND COMPOSITES

The material properties are as follows:
Tensile modulus of the cured CFRP laminate: E_{CF} = 48 GPa
Elastic modulus of the H100 PVC foam core: E_{CORE} = 0.13 Gpa.

Explain the nature of stress distribution in the above sandwich composite and compare it with a homogeneous beam.

For a CFRP sandwich beam subjected to three-point bending, determine:

- The maximum normal (flexural) stress and its location
- The maximum transverse shear stress and its location

COMPOSITES - MANUFACTURING

Question 3.1

Our vehicle is designed with stringent aerodynamic requirements, which requires the use of complex and highly curved surfaces. Producing critical components like the aeroshell and chassis, along with integration, is a task that is both time-sensitive and demands precision. While there are various methods for manufacturing composite structures, it is essential to identify the fastest and most efficient method.

- What are the various methods in which fiber-reinforced composites can be manufactured?
- Think about how a core material can be integrated into these manufacturing processes. Can any material be used as a core during these process? What considerations must be made while selecting the core with regards to manufacturing?
- Compare and choose the method which offers the most control and reliability over quality, which can be implemented by Team Agnirath.
- Every composite manufacturing method requires a mold to manufacture parts with curvature. What are the possible considerations that must be made during the manufacturing of the molds?

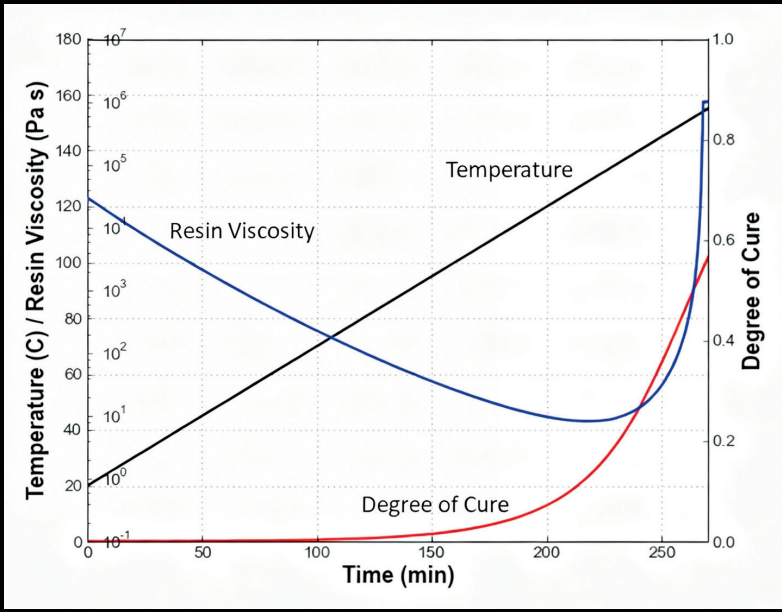
STRUCTURES AND COMPOSITES

AGNIRATH

Question 3.2

Vacuum Assisted Resin Transfer Molding (VARTM) is a manufacturing process where dry fibers called the preform, are laid over the mold while resin is pulled through into the preform assisted by a vacuum pressure gradient.

What are some key effects that must be kept in mind while performing VARTM? Can any combination of fiber reinforcement and resin matrix be used in VARTM? If there are any specifics list them. What are the constituents and materials used in VARTM and why?



- The characteristics of a particular resin system are given above. Notice that the scale of resin viscosity is logarithmic. What is the significance of gelation of the resin system, in the context of Composite Manufacturing?

- What are some process parameters that determine the quality of the component? In particular, What is the mathematical dependence of time taken on the pressure gradient and the viscosity of resin?

- We run simulations in order to analyze complex systems in a cost effective and efficient way. In the context of Resin Infusion, how can simulations be run, and what are the parameters to be provided? What could be the primary objective of simulating the process of Resin Infusion?

STRUCTURES AND COMPOSITES

AGNIRATH

Testing

Even though simulation software provides built-in material property values, we still need to validate it with the real world data. To do this, we prepare composite specimens and perform tests to determine key material properties. Now we need to perform a 3-point flexural (bending) test on the composite specimen.

Consider a composite laminate with the following stacking sequence with each ply 0.5mm thick: [0°/ 45°/ 0°/ 45°/ 0°/ 45°/ 0°/ 45°/ 0°/ 45°]

For what materials is it not possible to determine flexural properties according to ASTM D790, and what are the reasons for this? What is the difference between a 3-point and a 4-point bending test? Does the layup sequence affect the flexural properties?

An edgewise test was done on this according to ASTM D790 [Refer to the ASTM Paper in the resources folder]. The lab technician sets up the specimen with the following measurements:

- Specimen width = 5 mm
- Specimen depth = 8 mm
- Support span = 128 mm
- Overhang on each end = 11 mm
- Applied force = 70 kgf

The technician claims the setup is fully compliant with ASTM D790 and reports a flexural stress of 477.8 MPa.

- You are the quality engineer. Identify every violation, correct the setup, and recalculate the true flexural stress. Also state whether the technician's stress value is conservative, and why that matters practically. State and justify your assumptions properly. If you feel the depth is wrong, then take it as the maximum permissible value.

- With the final value you have obtained, you have been asked to reduce the flexural stress by 35%. What changes would you make and why?

- Using the above measurements and the given test data, calculate the flexural strength and modulus of elasticity along the lengthwise of the specimen. [Flexure test data link]

AERODYNAMICS

AGNIRATH

By now, you should already have a basic understanding of the energy audit of the car. The car gains energy through solar irradiance and loses energy through a bunch of mediums. One such medium is the aerodynamic drag force. Aerodynamics majorly affects our car's performance, so having a good aero might be the difference between everlasting glory and being forgotten forever!!

Q1 - BASIC AERODYNAMICS

Let's begin with the basics of aerodynamics!

A. What is an airfoil? How does an airfoil generate lift? What are the other forces that are being generated by the airfoil? Give a brief description of different geometrical parameters of an airfoil and how they affect overall performance.

B. What is the difference between drag and drag coefficient? What are the different types of drag a car can possibly have? Do we want to minimize this? If yes, how? And if no, why?

C. Now that you have a brief understanding about the geometrical parameters of an airfoil, it is now time for us to select an airfoil. Identify the important parameters and relationships that we look at while choosing an airfoil.

D. Our car Aagneya, has a streamlined body with a rounded nose, canopy protrusion, smooth upper surface, and a long tapered rear section, it has a bullet shape to be concise. Even though this shape is designed to keep flow attached, localized boundary layer separation may still occur at certain places in the car.

- What is boundary layer separation and why does it occur? Why do we want to avoid this while cruising?
- Identify the areas in the car where boundary layer separation can possibly occur and justify your answer.

AERODYNAMICS

AGNIRATH

Q2 - DRAG REDUCTION & CROSSWINDS

In Australia, we will encounter more than 50kmph of crosswinds and we need ways to handle that and maintain the stability of the car and keep it moving forward. Even a slight mess up can cause the car to sway sideways or roll over, which we obviously do not want.
In WSC25 and during testing of the car for SASOL26, the car had a lot of yaw at high speeds. This is a major issue and has to be fixed in our next car because this makes it very unsafe and might end up causing an accident.



Here is a picture of Aagneya for reference

A. For the car that we will be taking to WSC27, we have to make changes in the aeroshell mainly for stability and drag reduction. Looking at our current car, what are some feasible improvements/ changes that you think we can make? Explain each point in detail.

B. What is downforce? In this scenario of stabilizing the car during crosswinds, do we need downforce or lift? List situations where a car needs more lift or more downforce and explain why in each case.

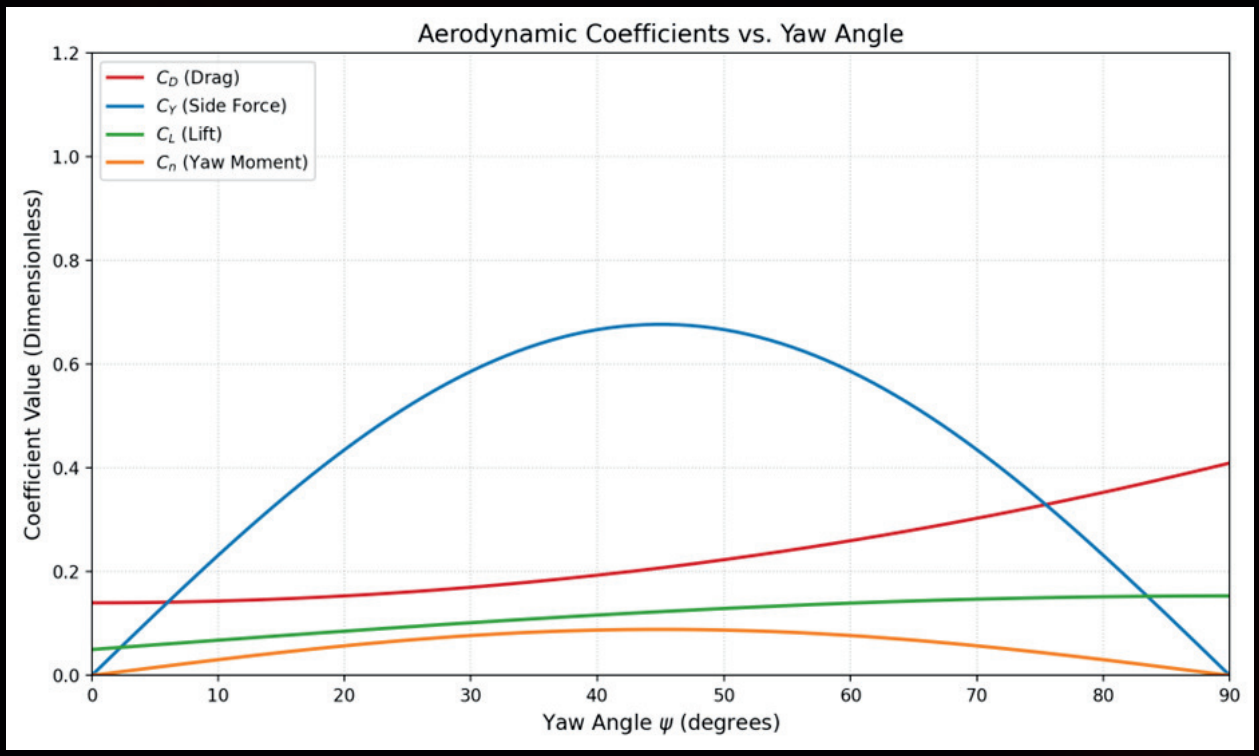
C. Okay now you have a grasp of downforce and lift and how they affect the car. Now, our car needs some lift to cruise properly. Estimate this value using necessary justifications. What is the trade-off between having high lift and high downforce, and what do you gain or lose with each?

AERODYNAMICS

AGNIRATH

Use the following data for the present car:
Mass with driver = 337 kg
Frontal area, A = 1 m^2
Lateral projected area, A_s = 1.35 m^2
Wheelbase, L = 2.80 m

Clearly state any assumptions that you're taking.



The graph given above shows how each coeffecient varies with the yaw angle.

Note: Yaw angle in this is defined as the angle between the x-axis and relative airflow.

Two aeroshell modifications are proposed:
Option A: Rear dorsal fin and smoother tail taper
This is intended to improve directional stability and reduce yaw in crosswinds.
ΔC_D = +0.006
ΔC_Y = -0.05
ΔC_L = -0.01
ΔC_n = -0.030

AERODYNAMICS

AGNIRATH

Option B: Rounded nose and reduced canopy side curvature
This is intended to reduce crosswind sensitivity and slightly lower drag.
$\Delta C_D = -0.004$
$\Delta C_Y = -0.08$
$\Delta C_L = +0.02$
$\Delta C_n = -0.012$

TASKS

- For the current car, calculate the apparent wind speed, drag force, side force, lift force, and yawing moment. Briefly explain why the car becomes unstable in crosswinds.

- For Option A and Option B, calculate the new values of C_D, C_Y, C_L, and C_n, and hence determine the new drag force, side force, lift force, and yawing moment. Compare both options with the current car and determine which option is better for WSC27 with proper reasoning.

Q3 - DESIGN AND SIMULATION (BONUS)

If you're entirely new to designing and CFD simulation, it's completely alright. Go through the attached references below to get a better understanding of the software tools. For this task, you are going to model any object of your choice and simulate it under certain conditions to find the lift and drag of the object. You are free to use any 3D modelling software and simulating software. If you don't know where to get started, I would highly recommend Fusion360 for 3D modelling and Ansys Fluent student version for CFD simulation.

- Fusion360
- Ansys Fluent Installation
- Ansys Fluent Tutorial
- Surface Modelling

AERODYNAMICS

AGNIRATH

3.1 - DESIGN

Sketch an object of your choice and create a surface model using that sketch. Export this sketch as a STEP file to be later used in Fluent. Remember – Make a surface model only, and not a solid model. The model has to have zero thickness! In Fluent, our model is going to act as a solid wall, so the insides of your sketch do not matter, only the outer boundary does, so model wisely. Don't forget to attach your CAD file along with your application as a .STEP file.

3.2 - SIMULATION

A. Now that you are done with design, let us move the fun part! For simplicity, let us only consider the fluid flow to be laminar.

Parameters:
Fluid – Air
Airflow – Laminar
Relative velocity – 50m/s

You are free to choose the other parameters to your likings. Mention whatever parameters you're considering along with a brief explanation on why you are doing so. Also, if you are taking any assumptions while setting up the sim, do mention the assumptions to ensure that we are on the same page as you while reviewing your application.

B. From the Sim, find the Lift and Drag forces generated by your object. Plot a pressure contour and flow vector direction of your fluid domain.

C. Realistically, if the object was given some thickness and weight, would it take off from the ground?

Note: Save the file as a .wbpz before starting your sim because ANSYS takes up a lot of RAM and there's a high chance your sim might crash.

COOLING

AGNIRATH

Q1 - BASIC COOLING SYSTEMS

1.1 - HVAC System

In automobile system, the HVAC system is essential for maintaining driver comfort, safety, and overall well-being. In extreme environments like the Australian Outback, where temperatures exceed 40 °C, an efficient and reliable ventilation system is essential. A key parameter used to evaluate the effectiveness of such a system is the *age of air*, indicates how efficiently fresh air circulates within the cabin.

- What is meant by the *age of air*. How can we measure or calculate the age of air effectively within the occupant compartment of a vehicle?
- During the race, the vehicle operates at a steady cruising speed of 65 km/h. Fresh air is supplied to the occupant cell through a front inlet called the NACA duct, resulting in an inlet mass flow rate of 0.0017 kg/s. The dimensions of the occupant compartment are 1350 mm × 490 mm × 550 mm. Using an air density of 1.225 kg/m^3, determine the *age of air* within the cabin.

1.2 - Cooling Techniques

Thermal management and Cooling is a critical aspect of electric vehicle design, especially in high-performance applications like a solar race car. Inadequate cooling can lead to overheating, thermal runaway, or even fire hazards.
A video clip HERE shows you just how dangerous it can be. A good understanding of thermal characteristics is therefore essential during the design phase.

- Identify the various cooling techniques used for battery packs in automobiles? Discuss the pros and cons of each method in detail. Based on the operational conditions and constraints of our solar car, which cooling approach would you recommend, and why?
- What are Phase change materials (PCMs), and how do they function in thermal management systems? Discuss both the benefits and potential drawbacks of using PCMs in automotive applications.

COOLING

AGNIRATH

1.3 - Critical Cooling Management

Efficient thermal management is critical for maintaining performance and reliability in both electric motors and solar panels.

- Identify and explain at least 2 - 3 viable cooling techniques for :
  (a) Motor Cooling of electric motors.
  (b) Solar Panel Cooling for photovoltaic panels

- Explain the principles of heat removal mechanisms of each method, and discuss how they relate with system's operation. Validate the effectiveness and reliability of these techniques.

Q2 - HEAT SINKS

A. What are heat sinks, and how do they facilitate thermal dissipation in electronic systems?

B. Explain the working principle of heat sinks. The performance of a heat sink is influenced by the nature of the airflow around it. How do factors in geometry, influence its cooling efficiency? Refer to Figure-1 for a sample heat sink design. What optimal design modifications can be made to its geometry to enhance cooling efficiency?
(Hint: Think in terms of the nature of airflow.)

C. As part of designing an efficient heat sink, clear understanding of fin performance parameters is essential

- What are *fin effectiveness*, *fin efficiency*, and *fin thermal resistance*? Explain their physical significance in the context of enhancing heat sinks
- State the governing equations for each of these parameters and identify the variables involved in the equation

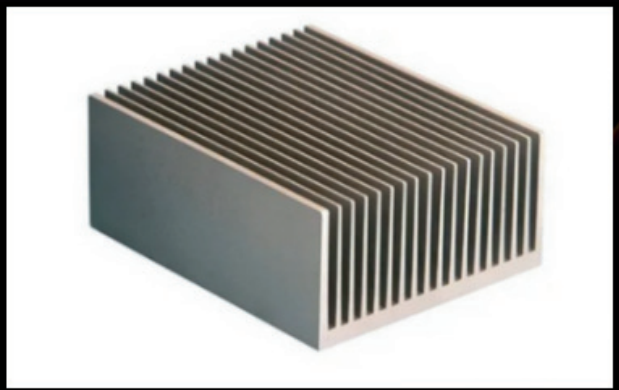


Figure - 1

Bonus: Design a CAD of optimized heat sink of dimension 75mm × 50mm × 40mm. You are free to choose the geometry, but the design should prioritize maximum heat dissipation and manufacturability. Submit your final model in . STEP format. (*Use fusion360 or any other software for designing)*

Estimate the approx. fin efficiency and fin effectiveness of your design. Take all necessary assumptions. Briefly explain how your chosen geometry accounts to improved thermal performance

COOLING

AGNIRATH

Q3 - MPPTs COOLING

Maximum Power Point Trackers (MPPTs) are devices in solar controllers that extract maximum available power from PV solar panels, by tracking and matching, the optimal voltage and current *(shown in figure-2 and figure-3)*
The two key components are power electronic switches (MOSFETs) and controller unit. Among these, MOSFETs experience significant thermal stress due to high-frequency switching, leading to substantial heat generation.

It has been observed that the efficiency decreases notably beyond 65°C, and the MPPT shuts down at approximately 70°C. Hence, maintaining the operating temperature within a safe range of 60 - 65°C is critical for reliable performance.

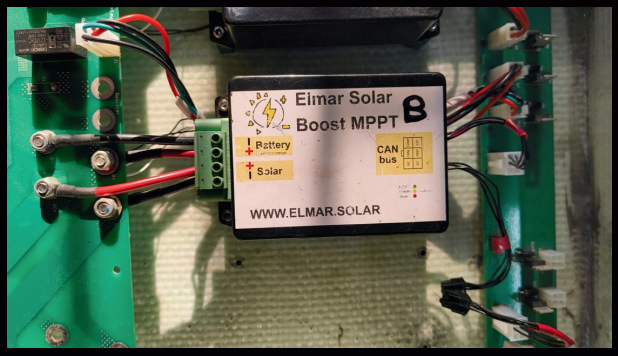


Figure-2 MPPT box

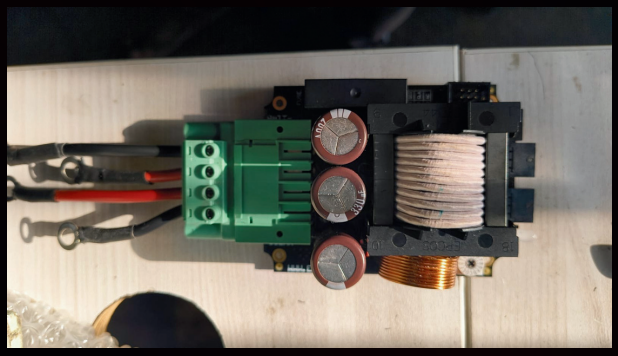


Figure-3 inside of the MPPT box when open

Given the above information, we shall estimate how much we can cool these mppts

3.1 - Cooling Strategies

- What other effective cooling techniques you can think implementing to maintain the MPPT temperature within the desired range?
  Compare all possible innovative methods, including:
  (a)Forced and passive air cooling
  (b)Phase Change Material (PCM)-based cooling.

- Discuss their feasibility, efficiency, and practical implementation constraints.

- What design strategies can be implemented to ensure uniform and effective airflow distribution across all MPPT units by any of the above methods?

COOLING

AGNIRATH

3.2 - Lumped Capacitance Modeling

We will use a method called “lumped capacitance“ to determine the temperature variation with time in a transient process. The essence of lumped capacitance method is the assumption that temperature inside solid is spatially uniform at any instant during a transient process.

The time taken for the mosfets to cool from a initial temperature of **70°C** to the desired safe operating temperature without any efficiency loss is **2 mins.**
The equation of lumped capacitance is given in equation.1

- Determine what the value of the convective heat transfer coefficient is from the above equation
- Find the velocity of air to maintain the MOSFET in the desired temperature range and to achieve the heat transfer coefficient by using Nusselt number
- Calculate the mass flow rate of air through the inlet duct of dimension 80 mm x 80mm

$$\rho V c_p \frac{dT}{dt} = \dot{E}_g - hA_s(T - T_\infty)$$

Equation.1

$$Nu = 0.023\, Re^{0.8}\, Pr^{0.4}$$

$$Re = \frac{\rho V D_h}{\mu}$$

$$Pr(Prandlt\ number\ of\ air) = 0.7$$

Use the above equations to calculate turbulent forced convection of air



** solar MPPT box*

COOLING

AGNIRATH

(Assume lumped capacitance is valid)

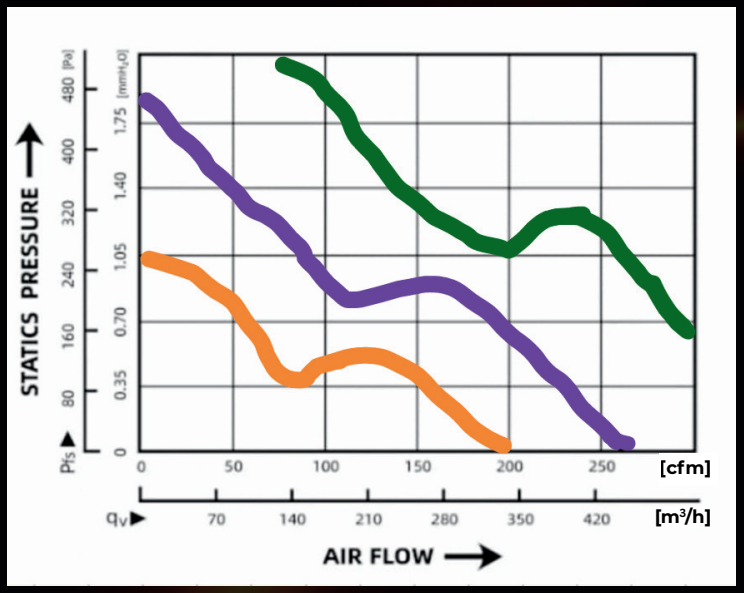
Data :

- Density of air = 1.225 kg/m³
- viscosity of air = 1.8 x 10^{-5} Pa . s
- Ambient temperature (T_∞) = 27°C
- Specific heat capacity (C_p) = 1.005 KJ/Kg .K
- Volume (V) = 0.375 m³
- Cross sectional area of thermal path (A_s) = 0.03125 m
- Thermal conductivity (K) of air = 0.024 W/m.K
- Heat generation (E_g) = 5.27 J
- D_h = 0.125 m
- characteristic length in air(L)= 50 mm

Q4 - FAN CURVES

- What is your understanding of use of fans used in cooling systems?
- What factors must be considered when selecting and configuring fans for the cooling system of our car ? How do you think they influence the cooling efficiency?
- Explain the property of a fan performance curve, including the relationship between airflow rate, static pressure, and operating point.
- What is meant by a master fan curve, and how is it experimentally or analytically determined for a given fan?

- From the air flow value (cfm) obtained from **question 3.2,** Identify the most suitable fan(s) from the given three fan curve plots by determining the operating point of the fan that can be used in the previous question



Bonus : Watch THIS fun youtube video and chill..

Note down key takeaways from this video and also what part interests you/ was new to you